Introduction

Purpose

This document specifies the requirements for CalvinFinder, a mobile app that helps users find classes, buildings and points of interest (POIs) around Calvin University. It defines what the system must be able to do and what qualities it must have so the team shares one vision for what the product should like before, during and after the development process.

Intended Audience

Developer team: Team G
Course Instructor: Prof. Victor Norman
Stakeholders: Team H
Future maintenance team

Intended Use

CalvinFinder is intended for anyone visiting or navigating Calvin University who needs to locate a building, room, or amenity. Typical uses would include an incoming student finding a classroom on the first day of classes, a Calvin guest trying to locate a building or anyone looking to find the nearest printer, bathrooms or vending machine. It is used to reference and help an individual find the way.

Product Scope

CalvinFinder is a mobile app that shows an interactive campus map, lets users select buildings to view their layouts, floors and displays POIs inside the buildings. Unlike other campus navigation tools, it combines classroom locations and POIs on one platform.

Definitions and Acronyms

Term	Definition
POI	Point of Interest such as vending machine, printer,water fountain, bathroom
Building Layout	A floor plan showing rooms and features in the building
Room	A number space in the building (e.g., SB 010 in Science Building)
SRS	Software Requirements Specification
UI	User Interface
MVP	Minimum Viable Product

FR/NFR	Functional / Non-functional requirement

Overall Description

Calvin Finder is a campus navigation app for Calvin University.

User Needs

Primary users: incoming students. These are newly admitted or first-year students who are enrolled members of the university but have little familiarity with the campus. They use the app to find classrooms, residence halls, dining locations, and offices, and to learn the layout over their first weeks. Getting lost makes them late, adds stress to an already stressful transition, and discourages them from asking for help.

Secondary users: visitors. These are external users with no formal role in the university, such as visiting parents and prospective students touring campus. They spend a short time on campus and use the app to reach specific destinations such as admissions, parking, and event venues without needing a guide. Getting lost costs them time and gives a poor first impression of the university.

Dependencies. Calvin Finder is a mobile app and depends on the following to function:

- **Mobile framework:** React Native, and the iOS and Android versions it supports.
- **Backend:** Supabase for the database and data storage, which requires an active Supabase project and network access to it.
- **Campus data:** Floor plans provided by Calvin University in PDF format. The team will manually convert these into nodes, connections, and room labels, so the app depends on the plans being accurate and on the team completing this conversion.
- **Third-party libraries:** React Native packages for map rendering and UI, and their continued compatibility with the framework version used.
- **Internet connectivity:** The app requires an internet connection (Wi-Fi or cellular data) to retrieve campus data and does not support offline use.
- **Reused components:** No software is reused from previous projects.

Assumptions.

- Users have a smartphone with an active internet connection while using the app.
- Calvin's floor plans are accurate at release, and campus changes are infrequent enough to be handled by manual updates.
- Supabase remains available and its usage limits are sufficient for the project's expected load.

Systems Functions and Requirements

Functional Requirements

FR-1 The user shall be able to view the campus map so that they can understand the campus layout.

FR-2 The user shall be able to pan and zoom the map so that they can look at specific areas in detail.

FR-3 The user shall be able to select a building on the map so that they can see its details.

FR-5 The user shall be able to recenter the map with one tap so that they do not get lost after panning.

FR-8 The user shall be able to select a building and view its floor layouts so that they can find locations inside the building.

FR-9 The user shall be able to switch between floors so that they can view different levels of a multi-story building.

FR-10 The user shall be able to view room numbers on floor layouts so that they can identify specific rooms.

FR-11 The user shall be able to view entrances, stairs, elevators, and restrooms on floor layouts so that they can orient themselves within a building.

FR-14 The user shall be able to view points of interest on the campus map so that they can find campus amenities.

FR-15 The user shall be able to select a point of interest so that they can view its name, type, and location details.

FR-16 The user shall be able to filter points of interest by category so that they only see locations relevant to their needs.

FR-17 The user shall be able to view points of interest within buildings on floor layouts so that they can find them indoors.

FR-18 The user shall be able to view the nearest point of interest of a selected type so that they can find it quickly.

FR-20 The user shall be able to search for buildings by name so that they can find them quickly.

FR-21 The user shall be able to search for rooms by room number so that they can locate specific rooms.

FR-22 The user shall be able to search for points of interest by category so that they can find desired amenities.

FR-23 The user shall be able to view search suggestions while typing so that they do not need exact spelling.

FR-24 The user shall be able to view recent searches so that they can repeat common lookups.

FR-25 The user shall receive suggested alternatives when no search results are found so that they can refine their search.

FR-26 The user shall be able to find locations using abbreviations or minor misspellings so that small mistakes do not prevent successful searches.

FR-27 The user shall be able to browse a list of buildings so that they can explore campus locations.

FR-28 The user shall be able to expand a building entry so that they can view its rooms.

FR-29 The user shall be able to select a location from the directory so that they can jump to that location.

FR-30 The user shall be able to drag the location panel so that they can balance map and directory visibility.

FR-32 Visitors shall be able to continue as guests so that they can use the application without a Calvin account.

FR-33 The user shall be able to reset a forgotten password so that they can regain access to their accounts.

FR-34 The user shall receive a clear error message when credentials are incorrect so that they know what needs to be corrected.

FR-36 The user shall be able to sign out so that their account remains secure on shared devices.

FR-37 The user shall be able to save a location as a favorite so that it can be accessed quickly later.

FR-38 The user shall be able to view favorite locations so that frequently used places are easy to access.

FR-39 The user shall be able to remove a favorite so that the favorites list remains relevant.

FR-40 Signed-in users shall be able to synchronize favorites across devices so that favorite locations remain available everywhere they sign in.

FR-41 The user shall be able to open application settings so that they can customize the application.

Non-Functional requirements

NFR-1 The campus map shall load within 3 seconds on a typical mobile connection.

NFR-2 Search results shall appear within 1 second of a user query.

NFR-3 Floor changes and POI filtering shall update within 1 second.

NFR-5 Passwords shall never be stored in plain text.

NFR-7 User sessions shall expire after a period of inactivity.

NFR-8 A first-time user shall be able to locate a building without requiring instructions.

NFR-9 The app shall display a loading indicator while data is being loaded.

NFR-11 The app shall allow access to previously loaded map and building data while offline.

NFR-12 The app shall display a clear message when network connectivity is unavailable or data cannot be loaded.

NFR-13 The app shall allow users to retry failed data requests without restarting the application.

NFR-15 Buildings, rooms, and points of interest shall provide meaningful labels for screen readers.

NFR-17 Information shall not rely solely on color for interpretation.

NFR-18 Interactive touch targets shall be large enough to support users with limited dexterity.

System Features
Campus Map, Building Layouts, POIs, Search and Browse, Accounts and Favorites, Error handling

Other Requirements -

Database will hold:

- Names of rooms
- POIs
- User account information (profiles, favorites, etc)

Relationships: A building has many floors, a Floor has many rooms, A POI belongs to one category and may belong to a Building or Floor. A user has many favorites.